

RidePulse

Campus Shuttle Tracking System

Real-Time Shuttle Tracking for Smarter Campus Travel

Project Overview

RidePulse is a campus shuttle tracking application that uses smartphone GPS and a backend system to provide students with live shuttle location, estimated arrival time, crowd level, and service alerts.

Smart • Real-Time • Campus Mobility

1. Problem Statement

Problem	Impact on Students
No live shuttle location	Students cannot know where the shuttle currently is.
Uncertain arrival time	Students may wait unnecessarily at shuttle stops.
Unknown crowd level	Students cannot plan whether a shuttle is comfortable to board.
Poor delay communication	Unexpected delays or route changes can waste time and affect class schedules.
Scattered shuttle information	There is no single platform providing current shuttle information.

2. Proposed Solution

Component	Input / Action	Output / Function
Driver Phone	Starts trip and shares location	Provides current GPS coordinates
GPS	Latitude + longitude + timestamp	Determines shuttle position
Backend / Firebase	Receives and stores live data	Processes location, status, ETA and alerts
Student App	Requests shuttle information	Displays map, ETA, crowd level and alerts
Admin	Manages routes, shuttles, drivers and alerts	Keeps the shuttle system updated

Step	Functionality
1	Driver logs in and starts a shuttle trip.
2	Driver phone obtains GPS coordinates at regular intervals.
3	GPS data is sent to the backend and the latest location is stored.
4	Student app receives the latest data and updates the shuttle marker.
5	System displays estimated arrival time for selected stops.
6	Crowd level and service alerts are displayed to students.

3. Key Features

Feature	Functionality
Live GPS Tracking	Displays the current shuttle position on a campus map using GPS data from the driver's smartphone.
Estimated Arrival Time (ETA)	Provides an approximate arrival time using shuttle position, route, distance and speed/average travel time.
Crowd Level	Shows Low, Medium or High crowd status. It can be updated by the driver or derived from passenger counts.
Route & Stop Information	Shows available shuttle routes and stops so students can identify the correct shuttle.
Service Alerts	Admin can publish delay, route-change, cancellation and other service announcements.
Push Notifications	Sends important updates such as delays and approaching-shuttle alerts.
Driver Module	Allows drivers to start/end trips, share GPS location and update shuttle status/crowd.
Admin Dashboard	Allows authorized administrators to manage shuttles, routes, drivers, users and alerts.

4. User Roles

User	Main Responsibilities
Student	Login, select route, view live shuttle location, check ETA and crowd level, and receive alerts.
Driver	Start/end trip, share GPS location, view assigned route and update shuttle status/crowd.
Admin	Manage users, shuttles, routes and drivers; monitor shuttle status; publish service alerts.

5. Crowd Level Logic

Level	Example Logic	Meaning
LOW	0-40% capacity	Plenty of space
MEDIUM	41-75% capacity	Moderately crowded
HIGH	76-100% capacity	Highly crowded / near capacity

6. Wireframe / Solution Flow

STUDENT HOME	LIVE MAP	SHUTTLE DETAILS
• Active shuttles • Route selection • Service alerts	• Campus map • ☐ Live shuttle marker • Stops & route	• ETA: 5 min • Crowd: Medium • Route status

End-to-End System Flow

Stage	System Action	Data / Result
1	Student logs in	Student dashboard becomes available
2	Driver starts trip	GPS tracking begins
3	Phone sends GPS coordinates	Latitude, longitude and timestamp reach backend
4	Backend processes data	Latest location, status and ETA information are maintained
5	Student app receives updates	Live shuttle marker and ETA are updated
6	Crowd / alerts are updated	Students see crowd status and service notifications

GPS Data Flow

Driver Phone	GPS	Backend / Firebase	Student App
Starts trip Location permission	Latitude Longitude Timestamp	Store + process Real-time update	Map + ETA Crowd + Alerts

7. Frontend & Backend Functionality

Frontend	Backend
Login and role-based screens	Authentication and user roles
Home dashboard	Shuttle, route and status data management
Live map	Real-time GPS storage and updates
ETA display	ETA calculation / route-distance processing
Crowd indicator	Passenger count or driver-updated crowd status
Alerts and notifications	Alert storage and notification delivery
Driver controls	Trip state and location update handling
Admin dashboard	Manage users, routes, shuttles and alerts

8. Suggested Technology Stack

Component	Technology
Mobile Frontend	Flutter / Android
Backend	Firebase or Node.js
Database	Firebase Realtime Database / Firestore
Authentication	Firebase Authentication
Location	Smartphone GPS
Map	Google Maps SDK/API
Notifications	Firebase Cloud Messaging

9. Expected Outcome

Area	Expected Benefit
Students	Less waiting and better travel planning using real-time information.
Transportation	Better visibility of shuttle movement, crowding and service status.
Administration	Centralized management of shuttles, routes, drivers and announcements.
Future Scope	Can be extended with analytics, demand prediction, route optimization and automated passenger counting.

RidePulse

Team Details

Team No.	16
Team Members	Akhilesh Sanvika Deekshitha Charvy Manish Akshaya

Less waiting • Better planning • Smarter campus transportation